

Workshop Title: CMMCA2026: The 5th Workshop on Computational Mathematics Modeling in Cancer Analysis

Workshop Date: Saturday, September 27th, 2026 (8:00 – 10:30)

Website: <https://cmmca.github.io/CMMCA2026/>

Brief Introduction: CMMCA 2026 brings together researchers and clinicians to advance mathematically grounded, interpretable approaches to multimodal cancer data analysis. This year's themes encompass spatial tumor ecosystems, generalizable multi-omics modeling for radiotherapy, and AI methods ranging from classical machine learning to foundation models. Contributions explore imaging reconstruction, cross-modal synthesis, tumor microenvironment analysis, biophysical simulation, and prognostic prediction, connecting computational innovations with biological understanding across scales to support precision oncology and address challenges in clinical translation.

Session 1 Title: Keynote Session

Session Chairs: Prof. Wenjian Qin, Prof. Jia Gu

Time	Paper Title
8:00–8:10	Open Remarks
8:10-8:30	Keynote: From Histology to Spatial Tumor Ecosystems: Multimodal AI for Cancer Analysis Prof. Tingying Peng, Helmholtz Munich
8:30-8:50	Keynote: Towards Generalizable Multi-Omics Modeling for Radiation Therapy Prof. Jing Cai, Hong Kong Polytechnic University
8:50-9:10	Keynote: AI in Cancer Research: From Classical Machine Learning to Foundation Models, and Beyond Prof. Yalin Zheng, University of Liverpool

Session 2 Title: Oral Session

Session Chairs: Prof. Ge Ren

Time	Paper Title
9:10-9:18	Oral 1 - Murari Amabti The Glioma Epigenome as a Potential Landscape: A Continuous G-CIMP Erosion Coordinate that the Methylation Class Discards
9:18-9:26	Oral 2 - Hongjie ZHU TriSCoV-Net: Cross-Scale Verified Virtual Immunomarker Proxy Generation from Diffusion Magnetic Resonance Imaging
9:26-9:34	Oral 3 - Wenyuan Huang Pulmonary Vascular Fractal Dimension on Non-Contrast Planning CT as a Potential Imaging Biomarker for Early Radiation-Induced Lung Injury: A Multi-Scale Structural Analysis
9:34-9:42	Oral 4 - Zhitong LI OncoRelay3D: Foundation-Model-Assisted Reconstruction of 3D Tumor Cell-Expression Fields for Mechanism-Aware Directed Cell-Cell Communication

9:42-9:50	Oral 5 – Chengyang LIU Free-breathing Time-resolved Cine MR Fingerprinting Using Self-supervised Motion Navigation for Liver Cancer Radiotherapy
9:50-9:58	Oral 6 - Marco. L Wittrich A Differentiable Brain Tumor Mass Effect Solver
9:58-10:00	Closing Remarks
10:00-10:30	Poster Presentation section & Tea break

Poster Presentation Section

1. ID-7: Nerve-Centric Heterogeneous Graph Learning for Tumor–Nerve Interaction Modeling in Survival Prediction

HanWenNan, Ying Cao, Bingxue Wang, Zhenghua Cai, Yingying Cao, Xiuyuan Zhang, Chunjun Qian, Luying Gui

2. ID-6: Bridging Slices to Volumes: Anatomy-Guided Latent Diffusion for 3D-Consistent MR-to-CT Synthesis in Radiotherapy

Yao Pu, Zhiqing Zhang, Peixin Yu, Mayang Zhao, Yitao Zhuang, Zhenxi Zhang, Jing Cai, Ge Ren

3. ID-5: D²TMIL: Disentangled Dual-Task Multiple Instance Learning for 1p/19q Status Prediction from Whole Slide Images

Xiangyu Zhao, Jiaxin Hou, Ranran Zhang, Hao Hu, Yanhua Lu, Ben Jin, Miaoxia He, Wenjian Qin

4. ID-4: Shared-Private Subspace Decomposition for Multi-Task Lung Cancer Classification in CT

Weiyi Chen, Chengcai Liu, Haolin Sang, Xingyu Huang, Chenghao Wang, Yi Wu, Jie Tian, Shuo Wang

5. ID-2: Zero-Shot Ultra-Quality 4D-MRI Motion Reconstruction Framework Based on Augmented Digital Phantom Pre-Training

Peilin Wang, Pengyuan Qi, Fan Zhang, Chenyang Liu, Lu Wang, Xiang Wang, Zhichu Li, Yimin Ni, Yinpeng Liao, Jie Deng, Zhiyong Yang, Zhaocheng Xie, Jing Cai, Tian Li

6. ID-19: A Differentiable Brain Tumor Mass Effect Solver

Marco L. Wittrich, Michal Balcerak, Lucas Zimmer, Ivan Ezhov, Bjoern Menze, Marie-Christin Metz, Benedikt Wiestler, Jonas Weidner

7. ID-14: Pulmonary Vascular Fractal Dimension on Non-Contrast Planning CT as a Potential Imaging Biomarker for Early Radiation-Induced Lung Injury: A Multi-Scale Structural Analysis

Wenyuan Huang, Lixin Ma, Jia Gu, Jiahui He, Xin Jiang, Wenjian Qin

8. ID-12: OncoRelay3D: Foundation-Model-Assisted Reconstruction of 3D Tumor Cell-Expression Fields for Mechanism-Aware Directed Cell-Cell Communication

Zhitong Li, Yixuan Jiang, Ziming Liu, Chenjie Wu, Ningfeng Que, Chao Li

9. ID-11: Free-breathing Time-resolved Cine MR Fingerprinting Using Self-supervised Motion Navigation for Liver Cancer Radiotherapy

Chenyang LIU, Xiang WANG, Lu Wang, peilin wang, Yimin NI, Fan Zhang, Victor Ho Fun Lee, Peng Cao, Tian Li, Jing Cai

10. ID-10: TriSCoV-Net: Cross-Scale Verified Virtual Immunomarker Proxy Generation from Diffusion Magnetic Resonance Imaging

Hongjie Zhu, Huayuan Zhou, Hao Tang, Hanyu Liu, Chao Li

11. ID-9: The Glioma Epigenome as a Potential Landscape: A Continuous G-CIMP Erosion Coordinate that the Methylation Class Discards

Murari Ambati